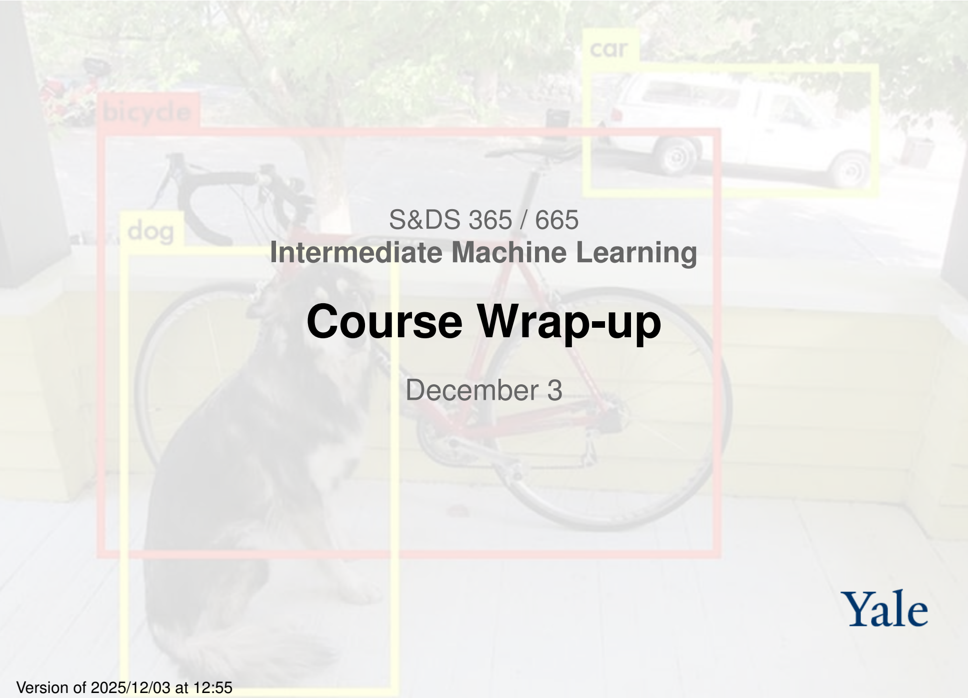


S&DS 365 / 665

Intermediate Machine Learning

Course Wrap-up

December 3

Yale

Version of 2025/12/03 at 12:55

For today

- New directions in ML/AI
- Differing views of our AI future
- Final exam logistics
- IML Survivor

But first: The usual reminders

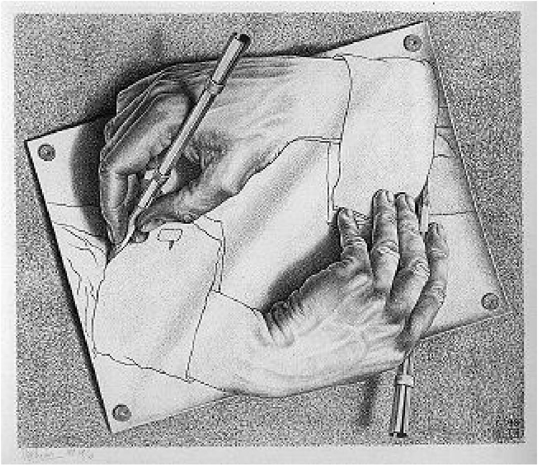
- Assn 5 due tomorrow
- Final exam: Friday Dec 12 at 9am, *HQ L02* (here!)
- Practice exams posted
- Review sessions:
  - Monday, December 8 at 5pm, KT 1101 (Zhehao)
  - Tuesday, December 9 at 5pm, Zoom (Andrea)
  - Wednesday, December 10 at 4pm, KT Mezzanine (John)

Rooms will be posted here: https://registrar.yale.edu/curriculum/final-exams

Frontiers of ML and AI

Some personal perspectives

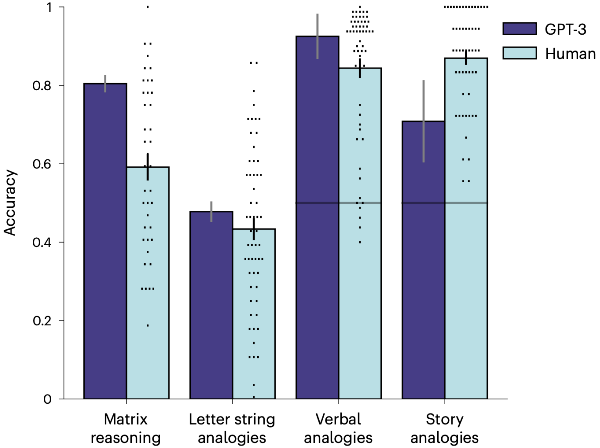
For today's discussion, recall the view that ML is the technological core that underpins AI systems. An analogy would be the distinction between nuclear physics and nuclear energy.



You scratch my back,
and I’ll scratch yours

$$[A, B] = AB - BA = 0$$

LLMs have emergent abstraction abilities



https://www.nature.com/articles/s41562-023-01659-w

But LLMs can fail catastrophically

"Two shapes are in a room. The shape next to the cube is blue. The shape next to the sphere is red. Which is red, the cube or the sphere?"

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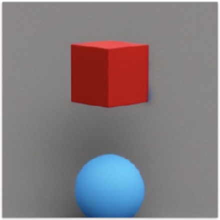

But LLMs can fail catastrophically

Based on the given information, it can be inferred that the sphere is red. The statement specifies that the shape next to the sphere is red, implying that the sphere itself is red. The color of the cube is not mentioned, so we cannot determine its color from the given information.

But LLMs can fail catastrophically

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Based on the given information, it can be inferred that the sphere is red. The statement specifies that the shape next to the sphere is red, implying that the sphere itself is red. The color of the cube is not mentioned, so we cannot determine its color from the given information.

Shortcomings are masked

- Innovations with Finetuning/RLHF hide these deficiencies
- System is trained to convince us
- Can lead to over-confidence and incorrect reasoning
- A central issue with “AI safety”

Another failure of relational reasoning

Two types of intelligence

1. "Sensory"— acquire semantic and procedural knowledge
2. "Relational"— rapid abstraction, association and generalization

Can both types be supported in a single architecture?
This is a frontier question for AI

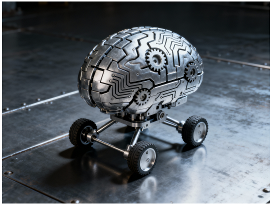
Supernatural intelligence?

What didn't evolution
discover wheels?

Supernatural intelligence?

What didn't evolution discover wheels?

Will AI evolve "cognitive wheels"?



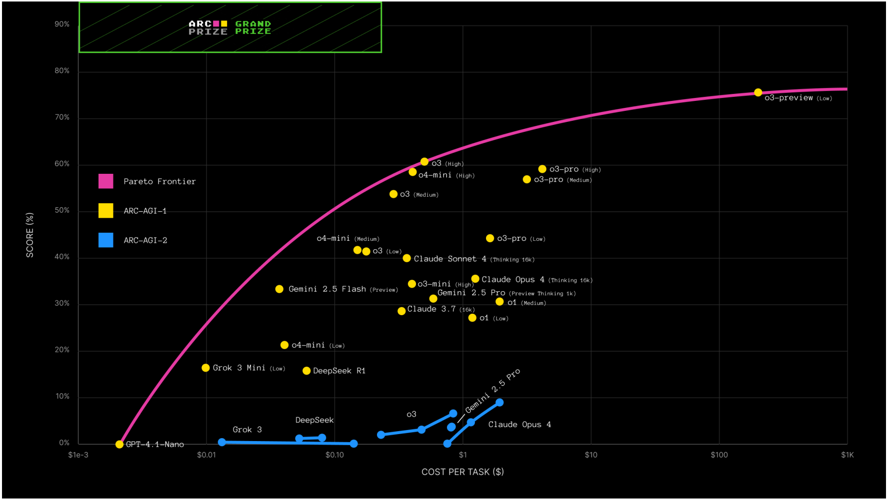
ARC-AGI Prize

arcprize.org; Chollet, “On the measure of intelligence” (2019)

ARC-AGI Prize

arcprize.org; Chollet, “On the measure of intelligence” (2019)

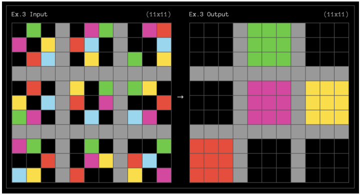
ARC-AGI Prize



arcprize.org; Chollet, “On the measure of intelligence” (2019)

“Platonic” abstractions

- What is required to discover *useful* “cognitive wheels”?
- Platonic view of primitive abstractions (small natural numbers, spatial relations) — they are real and exist independent of human thought



- Algorithms should be encouraged (through inductive biases?) to employ these abstractions

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Chollet, “On the measure of intelligence” (2019); Chollet, “OpenAI o3 Breakthrough High Score on ARC-AGI-Pub, 2024. `https://arcprize.org/blog/oai-o3-pub-breakthrough`, Beger et al. (2025) “Do AI models perform muman-like abstract reasoning across modalities?” arXiv:2510.02125

“Platonic” abstractions

“Using such abstractions is how we humans are able to understand our world, predict probable outcomes of our decisions, and to efficiently deal with novelty. These are all things we want AI systems to do in more trustworthy ways.”

Mitchell, “AI: A Guide for Thinking Humans,” October 2025; Beger et al. (2025) “Do AI models perform muman-like abstract reasoning across modalities?” arXiv:2510.02125

Creativity and rigorous reasoning

- We need new ML ideas to enable abstraction and analogy
- Humans often resist logical and evidence-based reasoning
- Can AI be based on more “rigorous” reasoning?

"We have Paleolithic emotions, medieval institutions and godlike technology. And it is terrifically dangerous, and it is now approaching a point of crisis overall."

Edward O. Wilson (2009)

AI and humanity: What do *you* think?

1. AI is just a new technology, like electricity, and civilization will advance much like it has for the last several hundred years; nothing will fundamentally change.
2. The world will fundamentally change, but humanity will find ways to safely control supernatural intelligent agents; AI will safely and peacefully improve the human condition.
3. AI poses a significant existential threat; humanity must learn to co-exist with supernatural intelligent agents or be wiped out by them.
4. Something completely different from the above three.

AI and you

1. AI will complement my future job, but I don't expect it to significantly change what I do in the future.

2. I need to reconsider my career goals and dreams, because AI is likely to dramatically change my planned career

We’ve covered a lot of ground!

Week	Dates	Topics	Demos & Tutorials	Lecture Slides	Readings & Notes	Assignments & Exams
1	Aug 27, Aug 29	Sparse regression	CO Python elements CO Pandas and regression CO Lasso example	Wed: Course overview Fri: Sparse regression	Google Colab Basics PML Section 11.4 Notes on linear regression	
2	Sep 3	Smoothing and kernels	CO Smoothing example CO Using different kernels	Wed: Lasso (continued) and smoothing	PML Sections 16.3, 17.1 Notes on computing the lasso	Quiz 1
3	Sep 8, 10	Density estimation and Mercer kernels	CO Density estimation demo CO Mercer kernels (1/3) CO Mercer kernels (2/3) CO Mercer kernels (3/3)	Mon: Smoothing and density estimation Wed: Mercer kernels	Risk bounds for local smoothing Notes on Mercer kernels	CO Assn 1 (updated) CO Updated Prob 1.1

We've covered a lot of ground!

4	Sep 15, 17	Neural networks and overparameterized models	example (1/2) np-complete example (2/2) TensorFlow playground	Mon: Neural networks Wed: Double descent	13.2 Notes on backpropagation Notes on double descent	Quiz 2
5	Sep 22, 24	Convolutional neural networks	Convolution demo CNN demo (1/2) CNN demo (2/2)	Mon: Convolutional neural networks Wed: CNNs and Gaussian Processes	PML Section 17.2 Notes on Bayesian inference Notes on nonparametric Bayes	Assn 1 in Assn 2 out ipynb converter
6	Sept 29, Oct 1	Gaussian processes and approximate inference	Parametric Bayes Gaussian processes Gibbs sampling for image denoising	Mon: Gaussian processes Wed: Recap of GPs Introduction to approximate inference	Notes on simulation	Quiz 3
7	Oct 6, 8	Variational inference	Variational autoencoders	Mon: Variational inference Wed: VAEs	PML Section 20.3 Notes on variational inference	Assn 2 in Assn 3 out ipynb converter
8	Oct 13	Midterm			Practice midterms	Oct 13: Midterm exam

We've covered a lot of ground!

9	Oct 20, 22	Graphs and structure learning	CO Graphical lasso demo	Mon: Sparsity and graphs Wed: Discrete data and graph neural nets	Notes on graphs and structure learning Graph neural networks PML Section 23.4	
10	Oct 27, Oct 29	Deep reinforcement learning	CO Q-learning demo CO DQN demo	Mon: Reinforcement learning Wed: Deep reinforcement learning	Sutton and Barto, Section 6.5	Assn 3 in CO Assn 4 out CO ipynb converter
11	Nov 3, 5	Policy methods	CO Policy gradients demo CO Actor-critic demo	Mon: Policy gradient methods Wed: Policy gradients (continued)	Sutton and Barto, Section 13.1-13.3, 13.5	Quiz 4

We’ve covered a lot of ground!

13	Nov 17, 19	Sequence-to-sequence models and Transformers	GPT-4 Python API	Mon: Sequence models and attention Wed: Transformers, LLM scaling PML Sections 15.4, 15.5		Quiz 5 Assn 4 in ipynb converter
14	Nov 24, 26	No class, Thanksgiving break				
15	Dec 1, 3	AI Societal issues	Transformer demo Minimal LLM decoder	Mon: LLM finetuning, postprocessing Wed: Course wrap up		Assn 5 in
17	Dec 12, 9am	Final exam			Practice exams	Registrar: final exam schedule

Final exam

- Length: About 1.5$\times$ midterm
- Cumulative, but emphasis on material after midterm
- Closed book, cheat-sheet (2-sided, handwritten)


OUTTHINK • OUTCODE
IML
SURVIVOR
OUTPREDICT

- Majority votes off a topic!
- But–a topic may have a hidden immunity idol! 
- What are you thinking? Persuade your classmates!

Vote a topic off the final!

1. Gaussian processes
2. Double descent
3. Reproducing kernel Hilbert spaces
4. Variational autoencoders
5. Graphical lasso
6. Deep Q-Learning
7. Policy gradient methods
8. GRUs

Advancing ML: Something for everyone

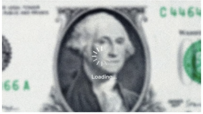
- Designing new methods
- Applying to new domains
- Solving deep mathematical puzzles
- Tackling unique engineering challenges
- Designing interfaces
- Collecting data
- Equitable use through policy and law
- Outreach and communication to broad communities

Always news

Science & technology

Robots can learn new actions faster thanks to AI techniques

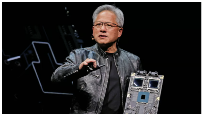
They could soon show their moves in settings from car factories to care homes



Finance & economics

Computers unleashed economic growth. Will artificial intelligence?

Two years after ChatGPT-3.5 arrived, progress has been slower than expected



Business

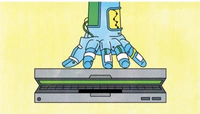
Nvidia's boss dismisses fears that AI has hit a wall

But it's "urgent" to get to the next level, Jensen Huang tells The Economist

Babbage

How AI will lead to a new scientific renaissance

Our podcast on science and technology. Nobel laureates Demis Hassabis and Jennifer Doudna, plus Google's James Manyika, present their vision for how AI is transforming science



Business

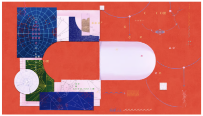
What ChatGPT's corporate victims have in common

The first casualties of generative AI offer lessons for other businesses

Science & technology in 2025

Supercomputers and AI are helping build better climate models

The coming year will be a big one for the advance of climate science



Science & technology in 2025

AI will boost drug development in 2025

Software-inspired medicines are getting closer to prime time

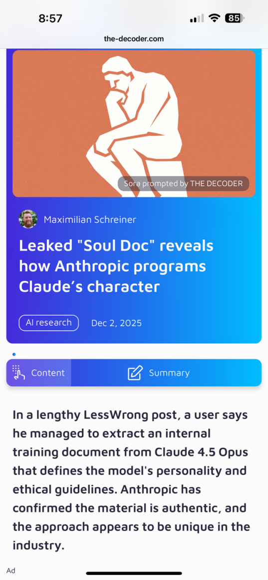
Finance in 2025

There will be no immediate productivity boost from AI

Everyone calm down a little

Always news


8:56
thestack.technology
AI
A 'soul document' was found compressed in Claude Opus 4.5 model weights
Anthropic's staff philosopher confirms the discovery is based on a legitimate document used to train Claude's most recent model.
Kiera Fields
Dec 02, 2025 - 4 min read
Image credit: https://unsplash.com/@aoddeh


8:57
the-decoder.com
Sora prompted by THE DECODER
Maximilian Schreiner
Leaked "Soul Doc" reveals how Anthropic programs Claude's character
AI research
Dec 2, 2025
Content
Summary
In a lengthy LessWrong post, a user says he managed to extract an internal training document from Claude 4.5 Opus that defines the model's personality and ethical guidelines. Anthropic has confirmed the material is authentic, and the approach appears to be unique in the industry.
Ad

Nanochat: Join us!

nanochat / README.md

karpathy Small fixes to typos 4763ce6 · 3 weeks ago History

Preview Code Blame Raw

nanochat

> The best ChatGPT that $100 can buy.

This repo is a full-stack implementation of an LLM like ChatGPT in a single, clean, minimal, hackable, dependency-lite codebase. nanochat is designed to run on a single 8XH100 node via scripts like speedrun.sh, that run the entire pipeline start to end. This includes tokenization, pretraining, finetuning, evaluation, inference, and web serving over a simple UI so that you can talk to your own LLM just like ChatGPT. nanochat will become the capstone project of the course LLM101n being developed by Eureka Labs.

Talk to it

Your input

- Please complete a course review!
- I greatly value your comments and feedback
- Feel free to send me comments privately
- Let me know how you use and continue to learn ML!

Thank you!